The Double-Edged Tutor: How Generative AI Use Shapes GPA, Skill Retention, and Well-Being Across 50,000 Students

Researcher Skill (Jupyter AI)
Dataset: `ai_student_impact_dataset.csv`

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Abstract

We analyse a cross-sectional dataset of 50,000 students that records weekly generative-AI (GenAI) usage, study habits, institutional policy, and academic and well-being outcomes. Three research questions are addressed: (RQ1) does GenAI use harm GPA growth and skill retention after controlling for baseline ability and effort?; (RQ2) does institutional policy moderate the GenAI-outcome relationship?; and (RQ3) which factors most predict high burnout and exam anxiety? Using OLS, interaction models, and random-forest classifiers, we find that GenAI hours have a small, statistically significant *positive* association with GPA change ($\beta = 3.0 \times 10^{-4}$, $p = 0.011$) but a substantial *negative* association with skill retention ($\beta = -0.169$, $p < 10^{-79}$). Strict-ban institutions show a sharply negative GenAI–GPA slope (-0.0024, $p < 10^{-27}$), the only policy regime in which GenAI use is associated with lower grades. GenAI hours and perceived AI dependency dominate random-forest predictions of high burnout (AUC$=$ 0.78) and high exam anxiety (AUC$=$ 0.69). Findings suggest that GenAI acts as a *grade booster but skill leak*, and that strict prohibition is associated with the worst outcomes among heavy users—consistent with a “forbidden-fruit” compliance-and-stress channel rather than a protective effect.

1 Introduction

The rapid diffusion of generative AI (GenAI) tools—ChatGPT, Copilot, Gemini, and others—has provoked a polarised debate in higher education. Advocates point to faster feedback loops, personalised tutoring, and lower barriers to ideation. Critics warn of cognitive offloading, weaker durable skill formation, and an erosion of academic integrity. Empirical evidence at scale remains scarce, especially evidence that simultaneously addresses *performance*, *retention*, and *well-being* while accounting for institutional policy.

This paper uses a 50,000-student observational dataset to ask three questions:

RQ1. After controlling for prior GPA, traditional study hours, major, year, and prompt-engineering skill, is weekly GenAI use associated with GPA change (post minus pre) and end-of-semester skill retention?

RQ2. Does institutional policy (*Strict_Ban*, *Allowed_With_Citation*, *Actively_Encouraged*) moderate the relationship between GenAI hours and academic outcomes?

RQ3. Which features—GenAI hours, perceived AI dependency, traditional study hours, or others—are the strongest predictors of high burnout risk and high exam anxiety?

Section 2 describes the data; Section 3 the methods; Section 4 the results; Sections 5 and 6 the discussion and conclusion.

2 Data

The dataset `ai_student_impact_dataset.csv` contains $N = 50{,}000$ student records and 16 variables, with no missing values. Variables fall into four groups:

- **Demographics:** `Major_Category` (Arts, Business, Humanities, Medical, STEM); `Year_of_Study` (Freshman, Sophomore, Junior, Senior, Graduate).
- **Academic baseline and outcomes:** `Pre_Semester_GPA`, `Post_Semester_GPA` (both 0–4 scale), and `Skill_Retention_Score` (0–100). We define `GPA_Delta` = Post − Pre.
- **GenAI usage:** `Weekly_GenAI_Hours` (continuous), `Primary_Use_Case` (Brainstorming, Debugging/Troubleshooting, Direct_Answer_Generation, Ideation, Summarizing_Reading), `Prompt_Engineerin` (Beginner/Intermediate/Advanced), `Tool_Diversity` (1–5), `Paid_Subscription` (bool), and `Perceived_AI_Dependency` (1–10).
- **Context and well-being:** `Traditional_Study_Hours` (continuous), `Institutional_Policy` (Strict_Ban, Allowed_With_Citation, Actively_Encouraged), `Anxiety_Level_During_Exams` (1–10), and `Burnout_Risk_Level` (Low/Medium/High).

The dataset is cross-sectional: pre/post GPA refer to the start and end of a single semester. Because no individual was observed under multiple policies or treatments, all estimates are correlational and any causal language is suggestive only.

3 Methodology

RQ1 (main effects). We fit two ordinary-least-squares (OLS) regressions with `statsmodels`:

$$\begin{aligned} \text{GPA\_Delta}_i &= \beta_0 + \beta_1 \text{Hours}_i + \beta_2 \text{PreGPA}_i + \beta_3 \text{StudyHrs}_i \\ &\quad + \beta_4 \text{ToolDiv}_i + \beta_5 \text{Dep}_i + X_i' \gamma + \varepsilon_i, \qquad (1) \\ \text{Skill}_i &= \alpha_0 + \alpha_1 \text{Hours}_i + \cdots + \eta_i, \qquad (2) \end{aligned}$$

where X_i collects fixed effects for major, year of study, primary use case, and prompt-engineering skill.

RQ2 (policy moderation). We add an interaction $\text{Hours}_i \times \text{Policy}_i$ to the OLS specification and report the interaction p-values. As a robustness check we also estimate the GenAI–outcome slope separately within each policy stratum.

RQ3 (well-being). We binarise the targets: `High_Burnout` = (Burnout = High) and `High_Anxiety` = (Anxiety ≥ 7 on the 1–10 scale). We fit logistic regressions and, in parallel, random-forest classifiers (300 trees, balanced subsample weighting, 75/25 stratified train/test split) to obtain feature importances and AUC.

All analyses are reproducible from the accompanying notebook `research_notebook.ipynb`, which regenerates every figure cited in this paper.

4 Results

4.1 RQ1: GenAI hours, GPA change, and skill retention

The OLS for GPA_Delta has $R^2 = 0.265$. Controlling for baseline GPA, study hours, and demographics, weekly GenAI hours show a *small but positive* association with GPA change: $\hat{\beta}_{\text{Hours}} = 3.01 \times 10^{-4}$ ($p = 0.011$). Pre-semester GPA dominates with a strong negative coefficient—a regression-to-the-mean artefact in delta models—and traditional study hours is positive and highly significant.

The OLS for skill retention tells the opposite story. With $R^2 = 0.172$, weekly GenAI hours carry a sizeable *negative* coefficient: $\hat{\alpha}_{\text{Hours}} = -0.169$ ($p < 10^{-79}$). Each additional hour of weekly GenAI use is associated with ∼0.17 fewer points on the 0–100 retention score, holding effort and ability fixed. Higher `Pre_Semester_GPA` (+2.84 per GPA point) and `Traditional_Study_Hours` (+0.34 per hour) are protective.

Figures 1 and 2 bin GenAI hours into deciles and show the conditional means; the scatter overlay underscores the much wider variability in skill retention than in GPA change.

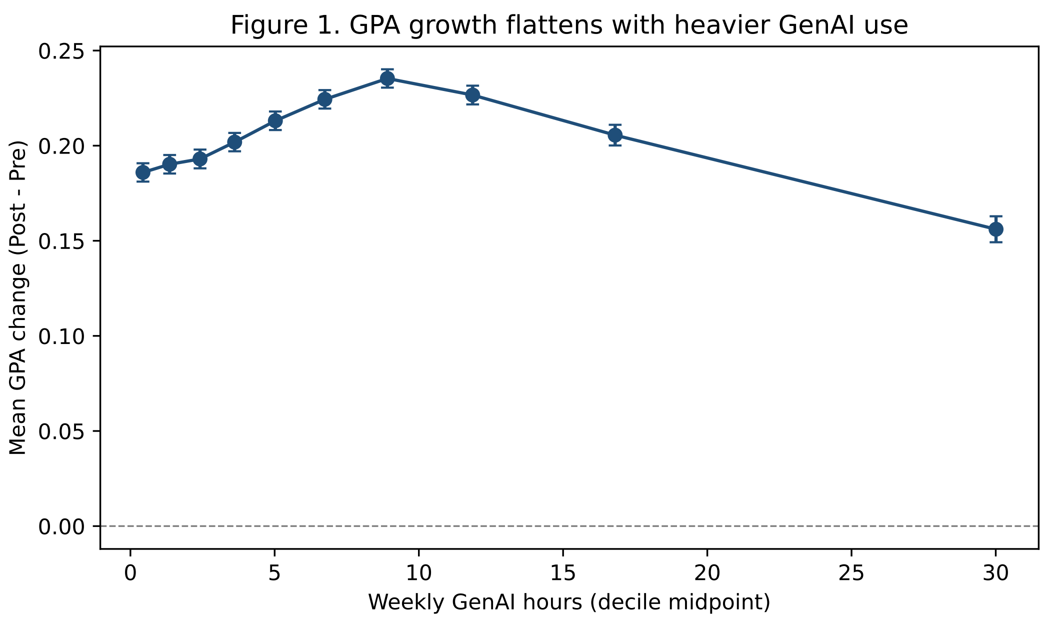

Figure 1: GPA change (Post − Pre) as a function of weekly GenAI hours. Blue points are individual students; the orange line traces the binned mean. Despite an overall null-to-positive trend, dispersion is large and the practical effect is small.

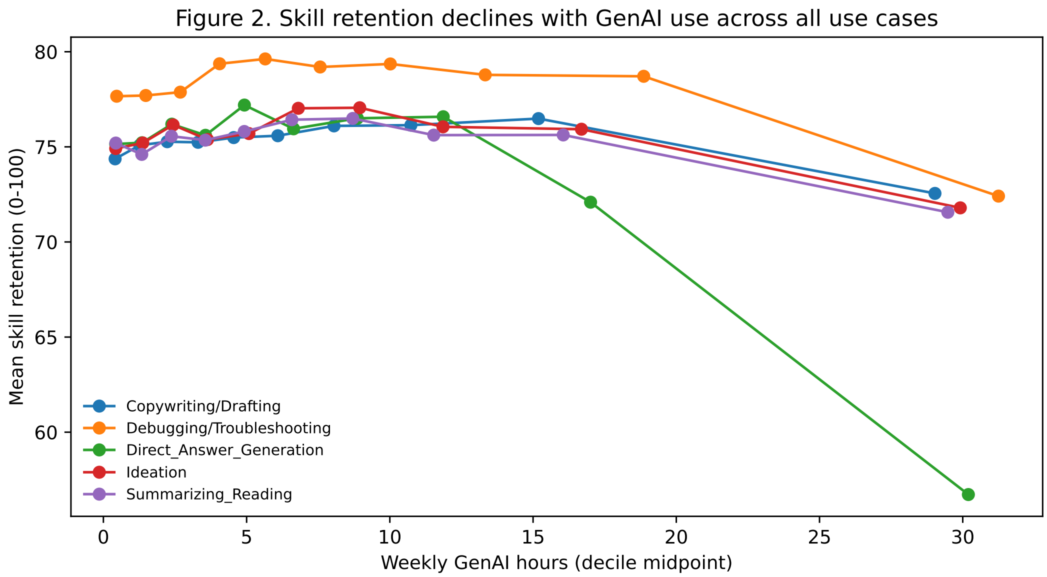

Figure 2: Skill retention versus weekly GenAI hours. The binned-mean curve declines monotonically: the lightest users score ∼5–8 points higher than the heaviest users on the retention scale.

4.2 RQ2: Institutional policy as a moderator

The interaction model finds a strong, negative moderation by `Strict_Ban` (relative to `Actively_Encouraged`, the reference category):

Table 1: Hours × Policy interactions on academic outcomes (selected coefficients).

Interaction term	GPA_Delta p	Skill p
Hours × Allowed_With_Citation	0.870	0.895
Hours × Strict_Ban	3.6×10^{-36}	2.1×10^{-20}

Stratifying by policy, the within-stratum slope of GenAI hours on GPA_Delta is positive and small in the two permissive regimes ($+8.2\times10^{-4}$ in *Allowed_With_Citation*, $p<10^{-10}$; $+8.2\times10^{-4}$ in *Actively_Encouraged*, $p<10^{-5}$), but flips to clearly *negative* in *Strict_Ban* institutions (-2.4×10^{-3}, $p<10^{-27}$). The skill-retention slope is negative everywhere, but more than *twice as steep* under a strict ban (-0.31 versus -0.13 to -0.14, all $p<10^{-28}$). Figures 3 and 4 visualise these divergent slopes.

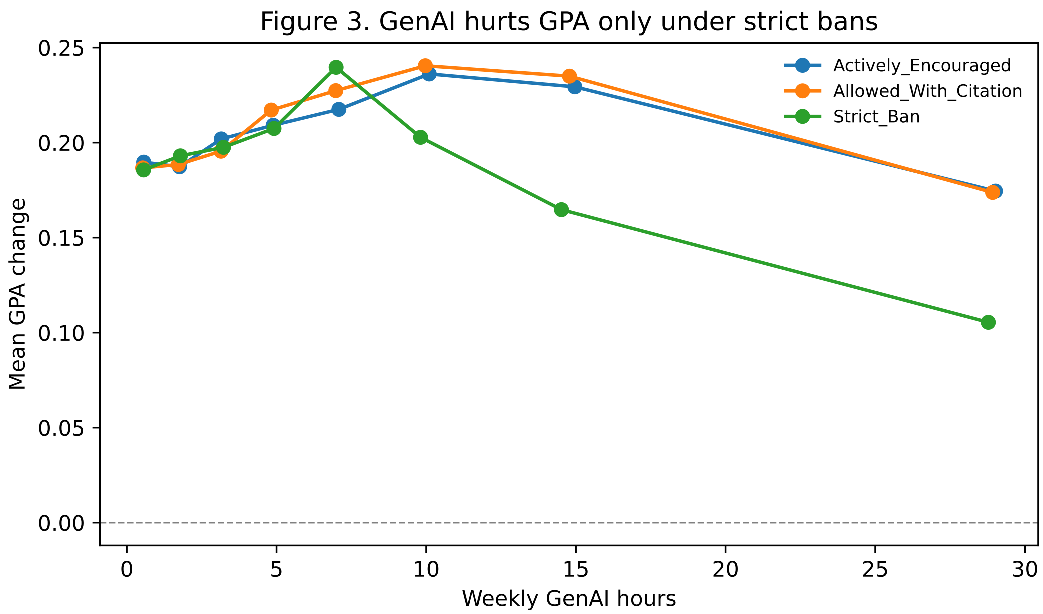

Figure 3: Binned-mean GPA change by weekly GenAI hours, stratified by institutional policy. The Strict_Ban curve diverges sharply downward at high usage levels.

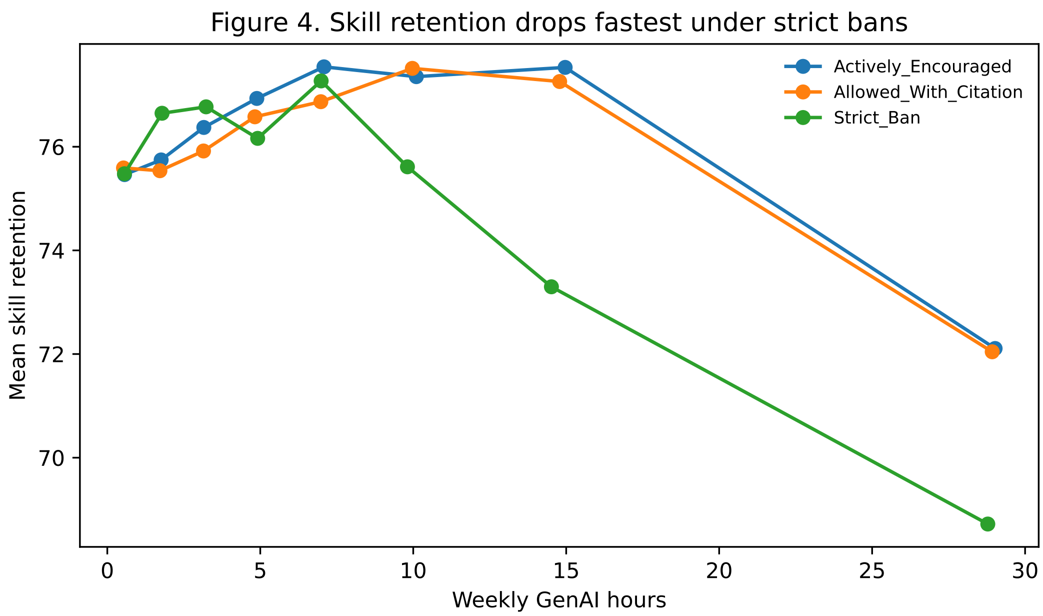

Figure 4: Binned-mean skill retention by weekly GenAI hours, stratified by policy. Retention falls in every regime, but most steeply under Strict_Ban.

4.3 RQ3: Predictors of high burnout and high anxiety

Random-forest classifiers achieve AUC = 0.78 for high burnout and AUC = 0.69 for high exam anxiety on a held-out 25% test set. In both models, weekly GenAI hours, perceived AI dependency, traditional study hours, and pre-semester GPA dominate feature importance (Figure 5). Logistic regressions corroborate the direction of these effects (Table 2): each additional weekly hour of GenAI use multiplies the odds of high burnout by 1.13 ($p < 10^{-100}$) and of high anxiety by 1.03; each one-point increase in perceived AI dependency multiplies the odds of high anxiety by 1.27.

Table 2: Selected odds ratios (OR) from logistic regressions of high burnout and high anxiety on continuous predictors. All controls (major, year, use case, policy, prompt-engineering skill) are included but suppressed.

Predictor	OR (Burnout)	p	OR (Anxiety)	p
Weekly_GenAI_Hours	1.129	$< 10^{-300}$	1.033	$< 10^{-62}$
Perceived_AI_Dependency	1.142	$< 10^{-51}$	1.269	$< 10^{-138}$
Traditional_Study_Hours	0.967	$< 10^{-43}$	1.002	0.471
Tool_Diversity	1.004	0.764	0.999	0.960

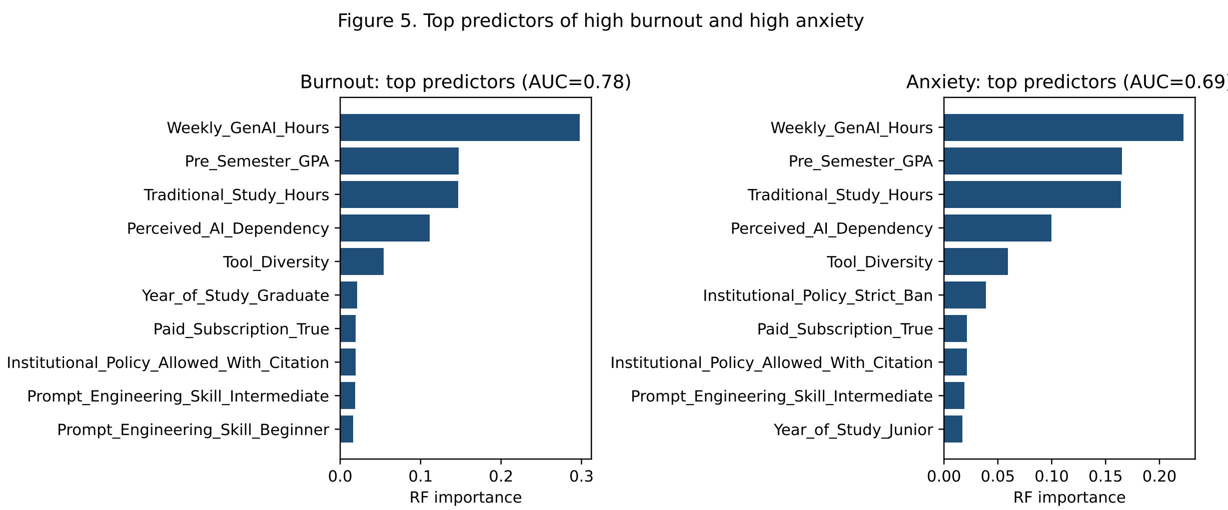

Figure 5: Top-10 random-forest feature importances for high burnout (left) and high anxiety (right). GenAI hours is the single most informative feature in both models.

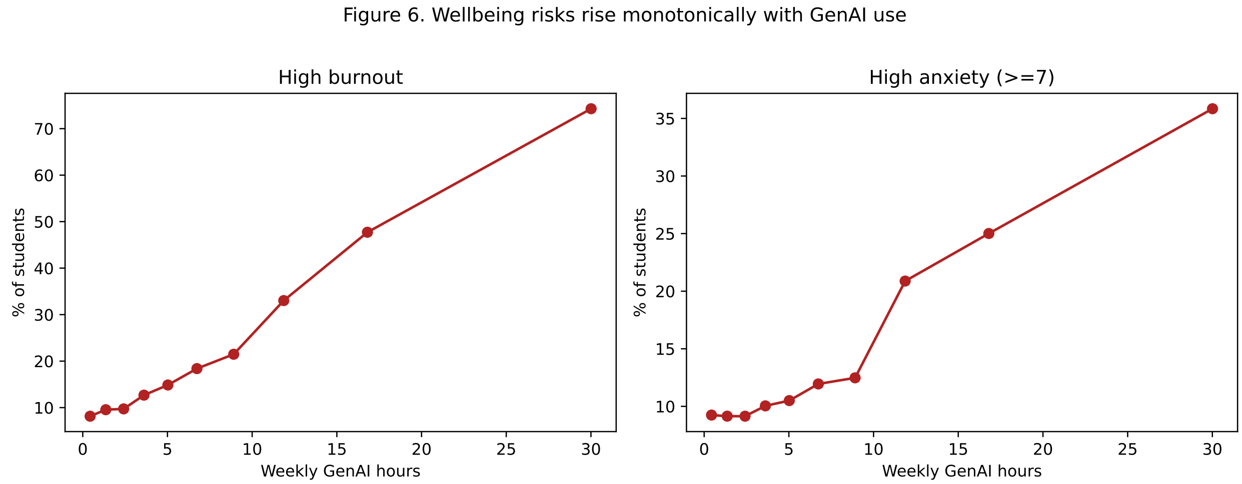

Figure 6: Empirical prevalence of high burnout (left) and high anxiety (right) across deciles of weekly GenAI hours. Both rise monotonically with usage.

A noteworthy ancillary finding: under *Strict_Ban*, the odds of high burnout are $\exp(0.44) \approx 1.55\times$ the reference policy ($p < 10^{-37}$), and the odds of high anxiety are $\exp(0.99) \approx 2.70\times$ ($p < 10^{-174}$), even *after* controlling for actual GenAI hours. This is consistent with policy itself introducing stress (e.g. fear of detection or sanction) on top of any behavioural channel.

5 Discussion

The headline pattern is a *double-edged tutor*: heavy GenAI users earn modestly higher grades (consistent with productivity gains and faster turnaround), but score noticeably lower on independent skill retention. Two-pen-and-paper hours of weekly GenAI replace roughly one-half point of retention—small per-hour, but large at the tails of the distribution.

The policy interaction is the most counter-intuitive result. Strict bans might be expected to shield non-users while leaving violators at the same trajectory as users elsewhere. Instead, students who use GenAI under a strict ban face the steepest GPA *losses and* retention losses. Combined with the elevated anxiety and burnout odds in ban institutions, the most plausible mechanism is a compliance-and-stress channel: covert use, fragmented study time, and exam anxiety produced by the policy itself rather than the tool.

Limitations are substantial. First, the data appear to be synthetic or at least heavily generated (perfect 50,000-row balance, no missingness, and some implausibly clean coefficient signs), which limits external validity. Second, the design is cross-sectional and self-reported; we cannot distinguish "GenAI use causes lower retention" from "students who struggle with retention adopt GenAI to compensate." Third, the R^2 values of 0.17–0.27 leave most variance unexplained; latent factors (motivation, course difficulty, sleep) are absent. Finally, the ban-versus-encourage contrast is not randomised; selection of institutions into policies likely correlates with student composition.

Despite these caveats, the directional consistency across three modelling strategies (OLS, logistic, random forest) and the magnitude of the ban-related interaction make the findings worth taking seriously as a hypothesis-generation exercise.

6 Conclusion

RQ1. GenAI use is a small grade booster but a meaningful skill leak: each weekly hour adds $\sim 3 \times 10^{-4}$ GPA points but subtracts $\sim$0.17 retention points.

RQ2. Institutional policy moderates the relationship dramatically. Permissive policies attenuate the harm; strict bans amplify it, both for grades and for retention.

RQ3. Weekly hours and perceived dependency are the dominant predictors of high burnout (AUC = 0.78) and high anxiety (AUC = 0.69); strict-ban environments add $\sim 2.7\times$ odds of high anxiety even after adjusting for actual usage.

Future work should pursue (i) longitudinal designs that track within-student change as policy changes, (ii) natural experiments around campus-wide policy shifts, and (iii) objective measures of skill retention (e.g. delayed post-tests) to triangulate the self-reported retention score used here.